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1 function xdot = modelBasic(t,y,pars, parsG, tst)
2 % Right hand side of the differential equations (from original code)
3
4 % Define volumes
5 Vpv = y(1);
6 Vlv = y(2);
7 Vsa = y(3);
8 Vsv = y(4);
9 Vrv = y(5);
10 Vpa = y(6);
11
12 % Define unstressed volumes
13 VpvU = parsG(1);
14 VlvU = parsG(2);
15 VsaU = parsG(3);
16 VsvU = parsG(4);
17 VrvU = parsG(5);
18 VpaU = parsG(6);
19
20 % Define resistances
21 RpuL = pars(1);
22 RmvaL = pars(2);
23 RavaL = pars(3);
24
25 Rsys = pars(4);
26 RtvaL = pars(5);
27 RpvaL = pars(6);
28
29 % Define compliances – elastance for right heart and left heart
30 Cpv = pars(7);
31 Csa = pars(8);
32 Csv = pars(9);
33 Cpa = pars(10);
34
35 % Define elastances (heart parameters)
36 ElvM = pars(11);
37 Elvm = pars(12);
38 ErvM = pars(13);
39 Ervm = pars(14);
40
41 % Define heart timing
42 TC = pars(15);
43 TR = pars(16);
44
45 % Left heart dynamics
46 Elv = Elastance(t-tst,ElvM, Elvm, TC, TR);
47 Plv = Elv*(Vlv-VlvU);
48
49 % Right heart dynamics
50 Erv = Elastance(t-tst, ErvM, Ervm, TC, TR);
51 Prv = Erv*(Vrv-VrvU);
52
53 Ppv = (Vpv-VpvU)/Cpv;
54 Psa = (Vsa-VsaU)/Csa;
55 Psv = (Vsv-VsvU)/Csv;

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56 Ppa = (Vpa-VpaU)/Cpa;
57
58 % flow into LH (mitral valve)
59 if Ppv > Plv
60     Qmval = (Ppv-Plv)/Rmval;
61 else
62     Qmval = 0;
63
64 end
65 % flow out of LH (aortic valve)
66 if Plv > Psa
67     Qaval = (Plv-Psa)/Raval;
68 else
69     Qaval = 0;
70 end
71
72 % flow into RH (tricuspid valve)
73 if Psv > Prv
74     Qtval = (Psv-Prv)/Rtval;
75 else
76     Qtval = 0;
77 end
78 % flow out of RH (pulmonary valve)
79 if Prv > Ppa
80     Qpval = (Prv-Ppa)/Rpval;
81 else
82     Qpval = 0;
83 end
84
85 %Calculate flows
86 Qpul = (Ppa - Ppv)/Rpul;
87 Qsys = (Psa - Psv)/Rsys;
88
89 % Differential Equations
90 dVpv = Qpul - Qmval;
91 dVlv = Qmval - Qaval;
92 dVsa = Qaval - Qsys;
93 dVsv = Qsys - Qtval;
94 dVrv = Qtval - Qpval;
95 dVpa = Qpval - Qpul;
96
97 % Right hand side of differential equations
98 xdot = [dVpv; dVlv; dVsa; dVsv; dVrv; dVpa];
99

```